DOI. No. 10.1109/MAES.2023.3334689

Tutorial:

An End-to-End Modular Framework for Radar Signal Processing: A Simulation-Based Tutorial

***Salman Liaquat* and *Nor Muzlifah Mahyuddin*, Universiti Sains Malaysia, Penang 14300, Malaysia**
***Ijaz Haider Naqvi*, Lahore University of Management Sciences, Lahore 54792, Pakistan**

This tutorial presents an end-to-end modular framework for signal processing techniques used in radar systems. The taxonomy of radar has been reviewed, as well as the subsystems of radar. The radar range equation is discussed, and its terms are explained. The radar operating principle and radar signal processing techniques to implement those operations have been highlighted. The simulation of a complete radar has been performed, and the results have been shown after each stage/module of a radar, enabling the reader to appreciate the impact of the specific process on the radar signal. The transmitter, antenna, and receiver subsystems have been simulated in MATLAB with a detailed discussion on waveform generation and transmission, receiver side operations such as matched filtering, Doppler processing, moving target indication filtering, and constant false alarm rate detection. Each step of the receiver chain's signal processing has been discussed, along with its influence on the signal. The implementation is carried out by simulating a generic 2D pulse-Doppler radar. A comprehensive analysis of the waveform generation and its effect on the target detection performance, varying radar cross section effect on the reflected wave, and path loss during propagation has been demonstrated. The sensitivity of the received signal to changing pulse types, number of fast Fourier transform points, window types, and CFAR algorithms have been presented. We have also discussed a radar system's conflicting range and Doppler ambiguities.

Authors' current addresses: Salman Liaquat and Nor Muzlifah Mahyuddin are with School of Electrical and Electronic Engineering, Universiti Sains Malaysia, Penang, 14300, Malaysia (e-mail: salmanliaquat@gmail.com; eemnmuzlifah@usm.my). Ijaz Haider Naqvi is with School of Electrical Engineering, Lahore University of Management Sciences, Lahore, 54792, Pakistan (e-mail: ijaznaqvi@lums.edu.pk). *(Corresponding author: Nor Muzlifah Mahyuddin.)*
Manuscript received 4 March 2022, revised 24 August 2022; accepted 16 November 2023, and ready for publication 20 November 2023.
Review handled by William Dale Blair.
0885-8985/23/$26.00 © 2023 IEEE

INTRODUCTION

Radio detection and ranging (radar) systems are used for detecting objects using electromagnetic waves (see Figure 1). They operate by transmitting a signal toward a target and receiving its reflected echoes [1]. Several target parameters, including range, velocity, and its orientation relative to the receiver, can be determined by analyzing these reflections through various radar signal processing (RSP) techniques. Radars can be used to identify aircrafts, ships, satellites, guided missiles, cars, weather patterns, and terrain [2]. Since its invention during World War II, radar's military and civil uses have expanded. The military has long used radar for aerial surveillance, navigation, detection, tracking, and weapons control and guidance. On the civilian side, radars have been used as tracking radars, anticollision radars, and ground penetrating radars for the detection of hidden objects. Moreover, there are weather radars and automobile speed detection radars. The medical industry has also started using them for various applications ranging from fall detection to vital signs monitoring [3], [4].

Three-dimensional (3D) radars compute height of the target in addition to the range, velocity, and azimuth information being computed by the two-dimensional (2D) radars. The phased array systems and beamforming techniques are the two fundamental aspects that enable the upgradation of 2D radars to 3D radars. Having phased array elements in a single dimension allows computation of target height whereas having a rectangular array can allow the radar to extract both elevation and bearing information of the target without mechanical rotation of the antenna. Phased array systems give several benefits over

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.



Image licensed by Ingram Publishing

single-antenna systems, including the ability to digitize the received signal at each antenna element, enabling the receiver to be modified and optimized according to their use [6], [7], [8], [9], [10], [11], [12], [13]. However, the complexity grows with an increase in the number of antenna elements. The number of elements in a phased array system affects the beam's directivity, half-power beamwidth (HPBW), and the array size. The beamforming can either be analog or digital. There are several electronically steered monopulse systems using analog beamformers suggested in the literature [11], [14], [15]. However, they are less flexible as compared to digital beamforming techniques implemented on the receiver side [16].

The problem of estimating the angles of arrival (AoA) using an array of antenna elements is of particular interest in radars to compute the height information of the target. Typically, analog beamforming radars employ a method called monopulse to measure angles, in which several simultaneous measurements are done with a single radar pulse [17]. Monopulse is an established approach for radars for precise angle estimation. A monopulse system for estimating one angle typically consists of two identical antennas, either separated by phase monopulse distance or at the same phase center but with a squint angle (amplitude monopulse), whose outputs are summed up to produce a sum beam and are subtracted to give the difference beam. If this antenna scans over a target, the sum beam output will be maximum when the target is exactly in the look direction, and the difference beam output will then be zero. Estimating the AoA of multiple sources from a single snapshot obtained by a set of noncoherent subarrays has been proposed in the literature. Substituting the phase estimators into the likelihood function makes it depend only on the AoA, which facilitates maximum likelihood estimation (MLE) of the AoA from a single snapshot [18]. The maximum likelihood (ML) concept reveals that the optimal angle estimate is derived by scanning the antenna over the target and using the maximum of this scan pattern as an estimate. Considerable work has been done in investigating several two-antenna systems (phase and amplitude monopulse, sum and difference beams, four quadrant monopulse antenna, etc.) since they were achieved in analog hardware. Most phased array radars of today directly apply the classical monopulse methods (e.g., the four quadrant monopulse antenna) to array antennas. It can be shown that a generic monopulse formula can be derived directly from the ML estimating process for all sorts of arrays [19]. Digital arrays, on the other hand, can

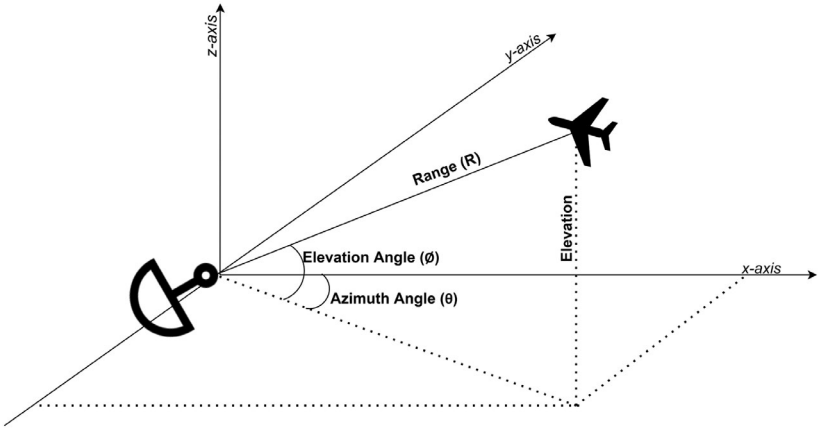


Figure 1.
Radar measurements [5].

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

employ either conventional monopulse approaches or innovative algorithms that take advantage of access to the element-level signals. Digital arrays, for instance, may use numerous simultaneous receive beams that are offset from a central beam in different directions. Some of these beams may be utilized just for angle measurements and not for detection. Using MLE methods, for instance, angle measurements are taken from numerous simultaneous receive beams [20].

With the advent of digital techniques and the availability of general-purpose computing hardware [21], [22], [23], [24], developing more versatile radar systems has become feasible, where signal processing algorithms are implemented in the software domain. The digital implementation allows radars to adaptively switch among the various modes of operation by modifying transmit waveforms and changing the RSP techniques on the receiver side. This approach also allows for upgradation of radar by simply upgrading to a new processing software capable of executing advanced RSP algorithms. For latest radar applications, having direct access to the data at each array element gives flexibility and real-time reconfigurability that was not possible with traditional systems, which need digital signal processing to begin after the formation of beams of a specified kind. In addition to this versatility, element-level digitization allows for the development of novel signal processing approaches [25].

However, developing radars for testing each case is not feasible. Therefore, simulation models for radars are beneficial and allow a deeper understanding of radar systems. Simulation techniques are used to design, evaluate, and optimize radar systems. It can save a lot of human and material resources that would otherwise be spent on assessing the performance of a specific radar system [26]. We begin this tutorial with a brief overview of the classification of radars, followed by an explanation of the operating principle of the radar and the radar signal processing (RSP) techniques. The subsystems involved in the operation of the radar are then discussed, followed by the scope of simulations and step-by-step results with analysis for appreciating the RSP techniques involved. The MATLAB code to reproduce the results of the section "Results and Discussion" can be found in the supplementary material, available online. This allows the interested reader to better comprehend the foundations of RSP by seeing the impact of modifying various variables in the code. Through a modular implementation of RSP algorithms, this article explains the working of the radar system, where each step's output is presented and analyzed. The goal is to put forth a modular framework for RSP by breaking down the radar system into subsystems and then sharing the output of each subsystem to give a complete understanding of its working. We can visualize the preprocessed and postprocessed signals at each step in order to better comprehend the process's underlying mathematical concepts.

TAXONOMY OF RADARS

Radars may be categorized in various ways, including classification according to their frequency, functionality, waveform, antenna location, type of beam steering, and AoA estimation approach, as illustrated in Figure 2. The radar considered in this tutorial for simulation is a *mechanically rotating S-band pulse-Doppler monostatic 2D radar*.

PRINCIPLE OF OPERATION AND RSP

In principle, a pulsed radar sends a train of radio frequency (RF) pulses, waits for the electromagnetic wave to be deflected from the target, and then directs the received signal toward the receiver. The pulsewidth τ is the time interval between a single pulse's rising and falling edges. The pulse repetition interval (PRI), denoted as T, is the period between pulses, the inverse of which is the pulse repetition frequency (PRF). The receiver amplifies the signal and determines the range of the target by determining the round-trip time taken by the pulse [5], given as

$$R = \frac{ct}{2} \tag{1}$$

where t is the time taken by the pulse to return (between 0 and T), c is the propagation velocity of the EM wave through the transmission medium, typically taken as 3×10^8 m/s.

The maximum unambiguous range $R_{\max}$ that can be measured by the radar is dependent on the PRF and is given by [5]

$$R_{\max} = \frac{cT}{2} = \frac{c}{2\mathrm{PRF}}. \tag{2}$$

The range resolution RR of a radar refers to its ability to discriminate between targets in close proximity in range at the same bearing and is given by [5]

$$RR = \frac{c\tau}{2}. \tag{3}$$

If two targets are within the range resolution of a radar, the radar system will treat them as a single target. The conventional radar also rotates about its azimuth axis, allowing computation of the target's azimuth angle, also known as bearing (θ), using the antenna direction measured from an antenna-rotation sensor.

RADAR RANGE EQUATION

The radar range equation, given by (4), calculates the power P_r returning to the receiving antenna (for a monostatic radar) [27], [28]. This equation demonstrates that

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

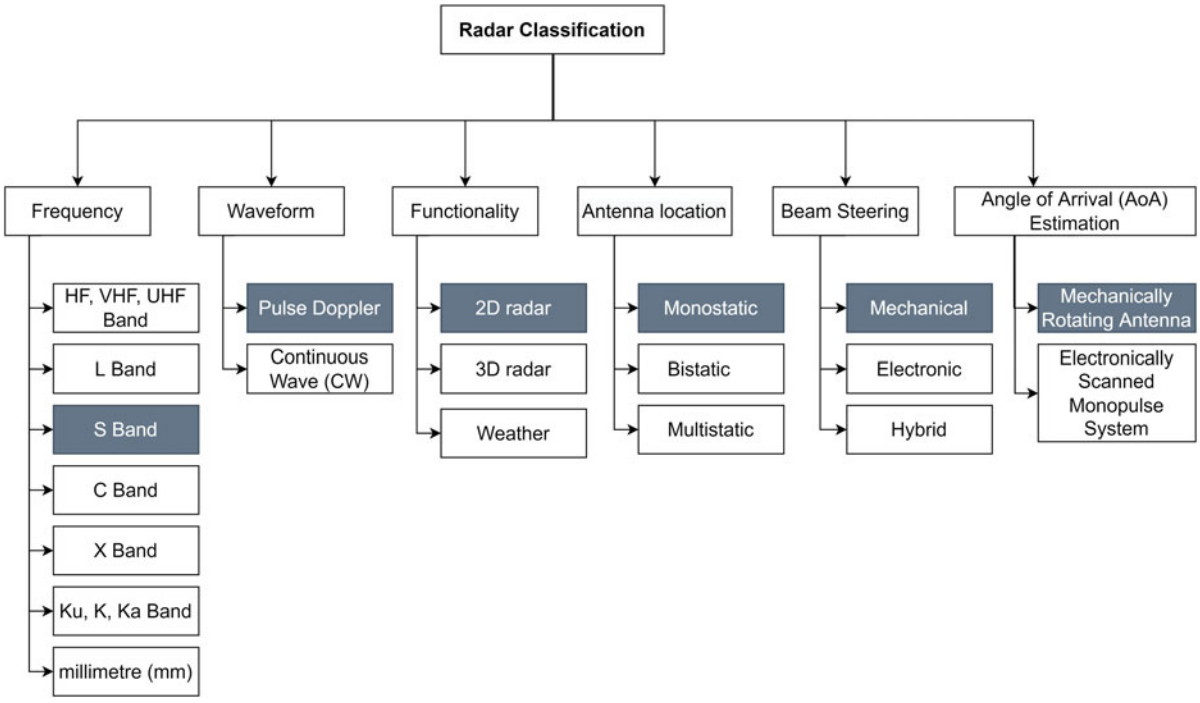


Figure 2.
Classification of radars [2].

received power decreases as the fourth power of range, implying that reflected power from distant targets is negligible compared to nearby targets. By neglecting the transmission and reception losses, it is expressed as [28]

$$P_r = \frac{P_t G_t A_r \sigma F^4}{(4\pi)^2 R^4} \tag{4}$$

where P_t is the transmitted power, G_t is the gain of the transmitting antenna, A_r represents effective aperture of the receiving antenna, σ is the RCS of the target, F is the pattern propagation factor, and R is the distance from the transmitter to the target. The radar uses the same antenna for transmission and reception, therefore the antenna aperture A_r can be expressed in terms of gain G_t as [29]

$$A_r = \frac{\lambda^2 G_t}{4\pi}. \tag{5}$$

Therefore, the radar range equation may alternatively also be written as

$$P_r = \frac{P_t {G_t}^2 \lambda^2 \sigma F^4}{(4\pi)^3 R^4}. \tag{6}$$

NOISE FLOOR

The ability of the radar to detect a target depends not only on the strength of the received signal, but also on the amount of noise present in the receiver. This noise is an inevitable component created by the thermal movement of conduction electrons in the resistive sections of the receiver input stages. The thermal noise power (N_t) in Watts at the antenna is given by [1]

$$N_t = kT_0B \tag{7}$$

where T_0 is the temperature of the ohmic parts of the circuit in Kelvin (K), B is the bandwidth in Hz, and k is the Boltzmann's constant, which has a value of 1.38×10^{-23} Joules per Kelvin (J/K). Much of the signal processing aims to mitigate or overcome this noise. The noise floor sets threshold in radar systems below which intended target's echoes are hidden as the radar cannot detect them. The observed signal must be greater than the noise floor for the receiver to detect it [1]. The noise floor (in dBm) can be determined using a noise figure (NF) in dB and is given by

$$\text{Noise Floor}_{dBm} = 10\log_{10}(1000kT_0) + NF + 10\log_{10}(B). \tag{8}$$

RADAR CROSS SECTION (RCS)

RCS is defined as the area intercepting the amount of power that produces the same received power in a radar if radiated isotropically [5]. It is the equivalent area observed by a radar and is mathematically given by

$$\text{RCS}(\sigma) = \lim_{R\to\infty} 4\pi R^2 \frac{|E^b|^2}{|E^t|^2} \tag{9}$$

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

where R is the target range, E^t is the transmitted power toward the target, and E^b is the back-scattered power toward the radar. RCS has units of area, and it depends on the frequency of operation, and target characteristics, such as the absorbance of the electromagnetic waves and the orientation of the target. Targets may be characterized as fluctuating or nonfluctuating depending on the RCS behavior. They can also be classified using Swerling Models. A nonfluctuating target has an RCS that is invariant in all directions, and is termed as Swerling-0 or Swerling-V target. Its probability density function is given by a Dirac delta distribution centered at the RCS value. A target is considered to be fluctuating when its RCS is not consistent, and it frequently fluctuates either in amplitude or phase based on its orientation over time. More realistic targets have Rayleigh (central chi-square of degree 2) or Rician (noncentral chi-square of degree 2) statistics, but even those are still just point targets. There are Swerling models for fluctuation targets, which are used in designing radar systems [30]. Higher fidelity target models use aspect dependent scattering models. For targets with RCS fluctuation statistics beyond the range of instances which are properly grouped by the Swerling cases have been discussed in literature, which includes the Weinstock cases [31].

MATCHED FILTERING AND PULSE COMPRESSION

Pulse compression is a signal processing technique primarily utilized in radar applications to simultaneously improve range resolution and signal-to-noise-ratio (SNR) by modulating the transmitted pulse and then correlating it with the received signal [32].

Modern radars employ LFM waveforms for target tracking, as the range-Doppler coupling enables better track accuracy in range [33]. Using LFM waveform allows equivalent sensitivity of a rectangular pulse, with a resolution improvement proportional to the bandwidth. LFM permits distinct control of the pulse energy through its duration and range resolution through its sweep bandwidth. The frequency of the LFM pulse increases/decreases at a constant rate throughout the period the pulse is on. The output of the matching filter is the autocorrelation function. Therefore to achieve both strong range resolution and good energy, a waveform with long duration and narrowly focused autocorrelation must be designed [5].

The instantaneous peak SNR at the receiver output can be obtained by matching the radar receiver transfer function to the received signal. The output of the matched filter has the maximum instantaneous SNR, range, and Doppler resolution [34]. Therefore, the matched filter receiver maximizes the return energy of the target's reflected pulse.

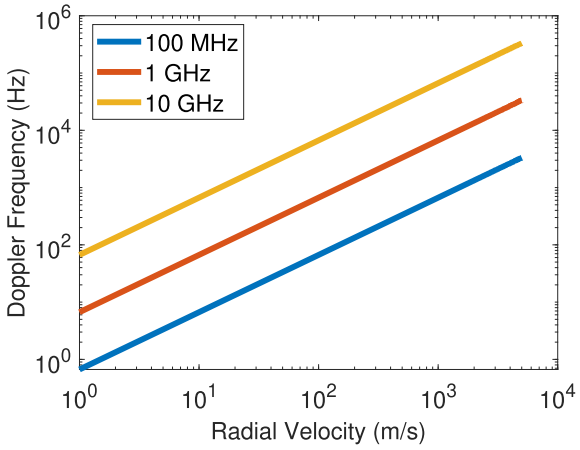


Figure 3.
Doppler frequency corresponding to radial velocities for varying frequencies.

DOPPLER PROCESSING

A Doppler shift is introduced in the signal received by the radar with the target's motion, altering the wavelength between the reflector (target) and the radar. The Doppler shift f_d for a target approaching toward the radar with a velocity $v_r \ll c$ is given by [5]

$$f_d = \text{PRF} = \frac{2v_r}{\lambda} \tag{10}$$

where v_r is the target radial velocity and λ is the wavelength of the transmitted signal. The Doppler shift is dependent on λ, the amount of Doppler shift changes with a change in the radar's operating frequency (see Figure 3). The same radial velocity corresponds to a higher Doppler shift for higher frequency of operation. The Doppler shift is proportional to the PRF and the unambiguous Doppler shift can be computed in the range $[\frac{-\text{PRF}}{2} \frac{\text{PRF}}{2}]$.

The Doppler shift may be positive or negative, depending on the direction of the target's velocity [34], with a positive Doppler for an approaching target and a negative Doppler for a target that is moving away from the radar.

Equations (2) and (10) show that the maximum unambiguous range is inversely proportional to the PRF, while the maximum unambiguous velocity is directly proportional to the PRF [5]. The higher the unambiguous velocity, the shorter the unambiguous radar range, and vice versa for a certain PRF. Low-PRF waveforms are often unambiguous in range but ambiguous in Doppler. In contrast, medium-PRF waveforms are ambiguous in both range and Doppler, while high-PRF waveforms are ambiguous in range but unambiguous in Doppler [1]. Therefore, PRF variation is used to resolve range and Doppler ambiguities in radar systems. A clustering technique for efficiently resolving Doppler ambiguity in dual PRF radars is

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

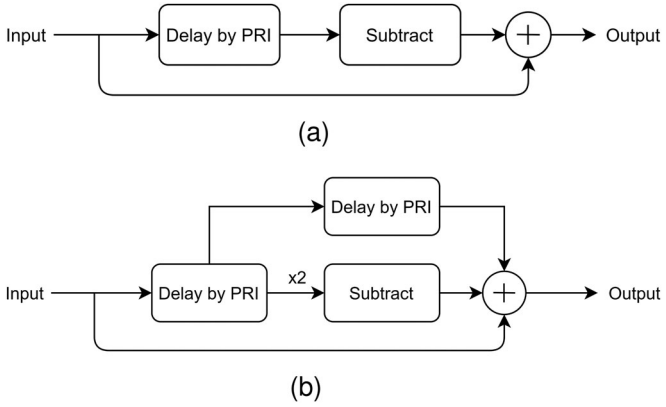


Figure 4.
MTI Cancellers. (a) Two-pulse canceller. (b) Three-pulse canceller.

presented in [35] when the number of PRF is insufficient to satisfy quantitative conditions for unambiguous Doppler resolution.

Range migration is a big challenge for pulse Doppler systems when the target velocity is fast compared to the range resolution and PRI. Range migration happens due to the long coherent processing interval, and the high radial velocity between the target and the radar along with the wide bandwidth, which causes significant linear range migration, which makes detection of weak targets difficult. Keystone transform is one of the methods to remove range migration of fast targets [36]. The Doppler Processing includes fast Fourier transform (FFT) and Windowing function which is explained here.

FFT AND WINDOWING FUNCTION

The slow-time axis contains the returns from consecutive identical pulses. For coherent radar systems, the phase difference between consecutive identical pulses can be used to measure the Doppler shift and estimate the target velocity. Moving targets in radars produce a time-varying spectral response, which can be captured by the FFT of the slow-time axis. FFT of this axis is used to convert time-domain samples to the frequency-domain samples. FFT-based Doppler processing isolates all targets according to their Doppler frequency, regardless of their range. It is a computationally efficient technique and produces satisfactory results for various signals. However, it has some limitations, such as poor frequency resolution. In addition, it causes the energy in a spectral response's main lobe to leak into the sidelobes, concealing and distorting other present spectral responses [37]. Side lobes are generated during signal processing, and sidelobe suppression is performed using window functions to minimize false alarms.

Windowing functions are implemented as finite impulse response (FIR) filters designed to give zero values outside of a specific interval [38]. The tradeoff between increasing the width of the main lobe and the sidelobe suppression capability of a window are essential factors to be considered while selecting the window function to be employed. A Dolph–Chebyshev window provides the best stop-band attenuation. However, its main lobe is the widest. As a result, it has superior noise reduction capability but less steepness. In contrast, the Hanning window offers a higher frequency selectivity with moderate main-lobe width and stop-band attenuation. As a result, the sidelobe attenuation for the Hanning window is not as significant as that of the Dolph–Chebyshev window [39]. The selection of a particular window depends on the radar application, which it is being designed for.

MOVING TARGET INDICATOR

MTI is a mode of operation for a radar that allows it to discriminate targets amid clutter [2]. An MTI filter minimizes clutter induced target-like returns while allowing returns from moving targets to pass through with little or no degradation. It exploits the fact that the target moves relative to the stationary clutter. For a particular sequence of radar pulses, the moving target's distance from the radar system changes; thus, the phase of the radar reflection returned by the target changes for subsequent pulses. In contrast, the stationary target/clutter results in the reflected pulses having the same phase shift. MTI is often implemented as a high-pass FIR filter in the slow-time axis.

Delay line canceller, which in reality is a radar filter that extracts the Doppler frequency shift and rejects the clutter frequency, is the defining property of the MTI radar. Some MTI radar filters can separate the moving target echo from a 70–90 dB greater clutter echo [40]. The delay line must introduce a time delay equal to the PRI. It can be implemented as a single-line or two-line delay canceller [34]. The clutter is present in the low-frequency regime and is therefore suppressed by the MTI filter [1]. From implementation perspective, a two-pulse canceller and a three-pulse canceller are shown in Figure 4(a) and (b), respectively.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

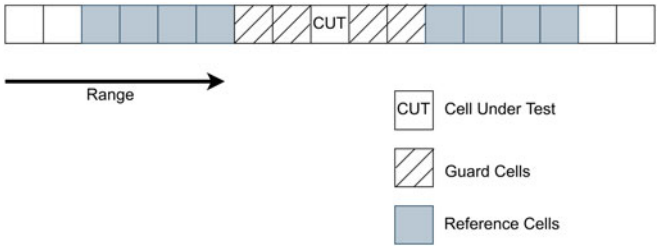


Figure 5.
One dimensional range window for CFAR [5].

CONSTANT FALSE ALARM RATE (CFAR) PROCESSING

CFAR detection is a standard adaptive algorithm used in radar systems to detect target returns in the presence of noise, clutter, and interference [41], [42]. The threshold level in the simple CFAR is determined by estimating the noise-floor level around the cell under test (CUT) as shown in Figure 5. It is calculated by averaging the power levels of a block of cells surrounding the CUT. Cells adjacent to the CUT, known as guard cells, are typically ignored to prevent this estimate from being tainted by the power from the CUT itself. A target is declared present in the CUT if the average power is greater than the sum powers of reference cells plus the local power level.

There are different types of CFAR processing, two of which are cell averaging (CA) CFAR and ordered statistic (OS) CFAR [4]. CA CFAR algorithm calculates the average value of the nearby cells. It is the most straightforward CFAR algorithm, but if a weak target is near a strong target, CA CFAR will mask the weak target, preventing it from being detected.

The CA CFAR implemented in this tutorial is based on the assumption of uniform clutter in the reference window. There are upgraded CA CFAR techniques allow for better identification of targets, particularly when the reference window comprises nonuniform clutter or transition zones from distinct clutter sources or signals from another target [4]. CA CFAR with the Greatest Of (GO) selection (CAGO CFAR) is more suited than conventional CA CFAR [43] when the clutter is not consistent in the reference window, or only the extremities of the clutter are contained in the reference window. On the other hand, when another target is present in one of the cells in the reference window, the detection threshold using CA CFAR becomes so high that it is difficult to identify the target in the CUT. In this scenario, CA CFAR with the Smallest Of (SO) selection (CASO CFAR) is proposed to recognize each of the two targets accurately [43]. However, when the clutter is relatively localized compared to the size of the reference window, the performances of both approaches become contrasted since, in this scenario, the strength of the clutter is overestimated, which might lead to its identification. Moreover, these solutions are predicated on assumptions of a static reference window. Thus, OS CFAR technique has been used for a more robust detection considering the numerous circumstances indicated above and its insensitivity to changes in environmental data [44].

In contrast to the CA CFAR approach, which is based on all the reference window samples, the OS CFAR method employs a single sample to estimate clutter plus noise power. The samples in the reference window are first sorted in ascending order, and the kth sample is selected to represent the average combined clutter and noise power. In the OS CFAR technique, it appears unnecessary to include guard cells because, in any event, the reference window samples are sorted before the clutter and noise power calculation [41]. Nevertheless, it should be mentioned that the use of a sorting algorithm makes the OS CFAR more complex to implement than other CFAR methods. A comparison on the performance of CA CFAR and OS CFAR has been presented in [4]. CFAR algorithms for various conditions have been presented in the literature [45], [46], [47], [48], [49], [50].

CFARs based on deep learning have also been proposed in the literature [46]. Convolutional neural networks (CNN) implementation with CFAR algorithms are becoming more prevalent [51]. Numerous researchers have suggested methods that combine CFAR and CNNs for the classification of targets ranging from microunmanned aircraft systems [52] to ship detection [53], [54], [55]. A multiclass CNN detector has also been proposed to replace the CFAR algorithm for identifying airborne objects for air traffic control radar [56].

SUBSYSTEMS OF 2D RADAR

Figure 6 shows a traditional 2D pulsed radar system. The subsystems and their functions are mentioned below.

1) **Transmitter**. The radar transmitter generates short, high-power RF pulses that are transmitted through the antenna. The transmitted waveforms may be rectangular or modulated. The modulated waveforms improve the radar performance as compared to the rectangular pulse, an example of which is a linear frequency modulated (LFM) waveform [57],

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

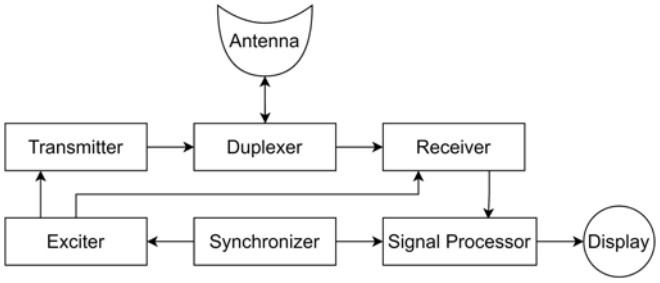


Figure 6.
Two-dimensional radar block diagram [1].

[58], [59]. Lower frequencies are preferred for surveillance applications, due to lower atmospheric loss and higher available powers [5]. However, this results in a larger wavelength and a bigger antenna.

2) **Duplexer**. The duplexer allows the antenna to switch between the transmitter and the receiver, thus allowing transmission and reception using the same antenna in a monostatic radar by providing the required isolation between the transmitter and receiver during the transmission operation. The high-power pulses would destroy the sensitive receiver without this switching. The duplexer functions like a circulator, shutting off the receiver circuitry during transmission and activating it to receive the signal returns once the transmission is complete [1]. Electronically active duplexers have also been implemented to isolate the transmitter and the receiver [60].

3) **Antenna**. The antenna distributes and efficiently converts the transmitter's energy to signals in space. This procedure is carried out identically at reception [1]. An antenna's most critical performance parameter is its gain, which is linked to directivity with an antenna efficiency factor [61], [62].

4) **Receiver**. The receiver amplifies and demodulates the received RF signals and outputs video signals.

5) **Radar Signal Processor**. The radar signal processor applies RSP techniques on the received pulses, including matched filtering, moving target indication (MTI), and Doppler processing. It implements numerous algorithms for target recognition and clutter filtering employed in modern radars. The RSP techniques employed on the receiver side determine the radar's performance.

6) **Display**. The display indication provides a continuous, clearly intelligible graphic representation of the targets' relative position to the radar.

7) **Synchronizer**. The synchronizer generates all triggers, i.e., timing pulses required to activate the transmitter, display sweep circuits, and ranging circuits.

SCOPE OF SIMULATIONS

In this tutorial, we have simulated the following subset of radar systems for an S-band 2D monostatic radar as shown in Figure 7.

1) **Transmitter**. The transmitter simulation includes the radar waveform generator, amplifier, and pulse transmission.

2) **Channel**. The transmitter pulse passes through the channel that attenuates the signal because of the path loss and adds noise to the pulse amplitude. The path loss is the reduction in signal strength caused by a line-of-sight (LoS) path through space with no impediments to induce reflection or diffraction and is dependent on the operating frequency.

3) **Targets**. Five targets having different ranges, velocities, and RCS are simulated for understanding various operations involved in radar and the subsequent effect of each RSP step. The targets delay the pulse and add Doppler shift depending upon their respective velocity. Moreover, the returning signal strength

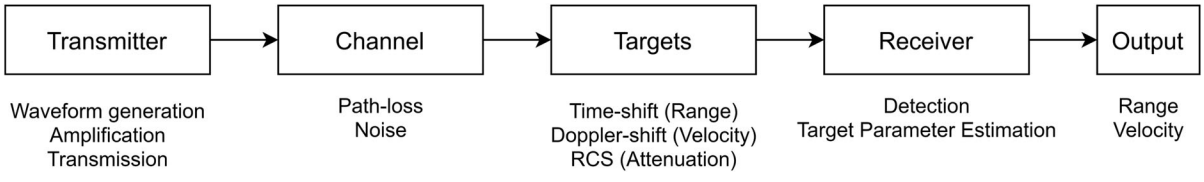


Figure 7.
Scope of simulation.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

Table 1.

Simulation Top-Level Radar Parameters	
Parameter	Value
Frequency	3 GHz
Pulse width	20 μs
PRF	1 kHz
Sampling rate	1 MSamples/s
Inter-Pulse Modulation	LFM (400 kHz BW)
Pulses to Integrate	20

depends on the targets' RCS and the distance between the target and the radar due to path loss.

4) **Receiver**. The receiver captures the reflections/echoes and performs RSP on these returns to extract targets and estimate their parameters. At the receiver, pulse compression, matched filtering, Doppler processing, and CFAR detection have been carried out.

5) **Output**. The range and velocity of the targets determined after applying the RSP operations are displayed as an output.

TOP-LEVEL PARAMETERS

The top-level parameters for implementation are mentioned in Table 1.

The wavelength λ is 0.1 m for this simulation as the frequency is 3 GHz, range resolution is 3 km if a rectangular pulse is used, the maximum unambiguous range is 150 km, and the unambiguous Doppler velocity range is 50 m/s (from -25 to 25 m/s). An LFM pulse of 400 kHz BW is being used in the simulated radar system.

CHANNEL AND ANTENNA SIMULATION

A simple channel that adds noise and attenuates the signal because of the path loss is simulated. The details of the noise floor at the receiver and path loss assumption and antenna gain are given in Table 2.

TARGET SIMULATION

We have simulated five targets with parameters shown in Table 3. Target simulation modifies the transmitted pulses to mimic target range, velocity, and RCS using time delay, Doppler shift, and attenuation, respectively.

Two targets (Target-1 and Target-2) have been placed within the range resolution capability of a rectangular pulse to observe the effect of employing an LFM waveform to distinguish two targets within the range resolution of the radar. At an 80-km range, one target (Target-3) is stationary to observe the effect of MTI in Doppler processing; Target-4 is a typical target to observe the effects of various signal processing on the received target echo. One target (Target-5) is positioned at a range of 145 km to determine whether the radar can detect a target close to its theoretical limit if the returning signal strength is adequate to cross the detection threshold within the radar's maximum unambiguous range. The targets scenario is depicted in Figure 8.

RECEIVER SIMULATION

The RSP steps implemented in the receiver simulation are shown in Figure 9. A matched filter is employed at the receiver front-end, followed by MTI filtering operation, Doppler processing to detect the velocity, and CFAR based detection to detect signals above a certain threshold. The parameters detected by the radar will be displayed at the output console.

COMPARISON OF RSP TECHNIQUES

A comparison of different signal processing is provided where the effects of changing the pulse type, the number

Table 2.

Channel and Antenna Simulation		
Parameter	Value	Remarks
Noise	-173 dBm	Minimum Noise Floor
Path loss	Dependent on $1/R^4$	Dependent on wavelength
Gain	151 dBi	Antenna and amplifier gain

Table 3.

Simulated Target Parameters			
	Range	Velocity	Remarks
Target-1	45 km	15 m/s	Overlapping Targets
Target-2	47.9 km	-15 m/s	Negative Velocity
Target-3	80 km	0 m/s	Static Target
Target-4	105 km	10 m/s	Normal Target
Target-5	140 km	20 m/s	Check range of radar

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

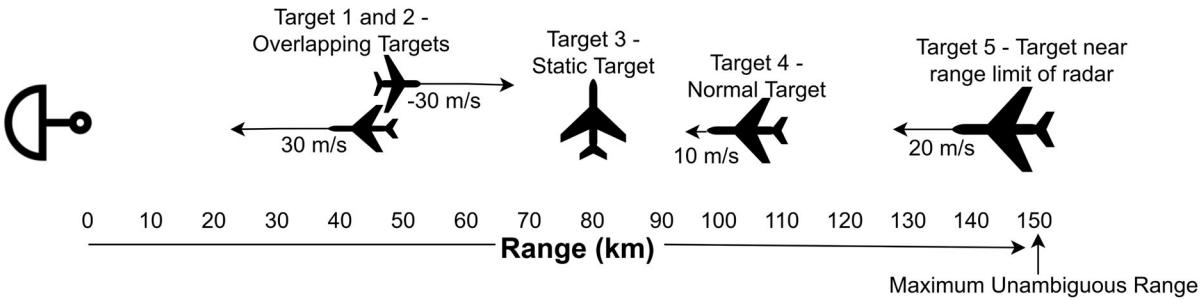


Figure 8.
Visual depiction of simulated targets.



Figure 9.
Scope of receiver simulation.

of FFT points for computing the Doppler shift, the windows, the MTI Filtering, and the CFAR algorithm on the output signal are discussed. These comparisons are followed by displaying Doppler ambiguities and the effect of RCS on the returning signal strength.

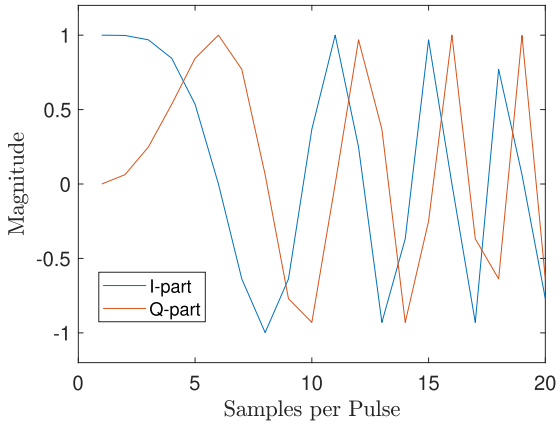


Figure 10.
I and Q parts of generated pulse.

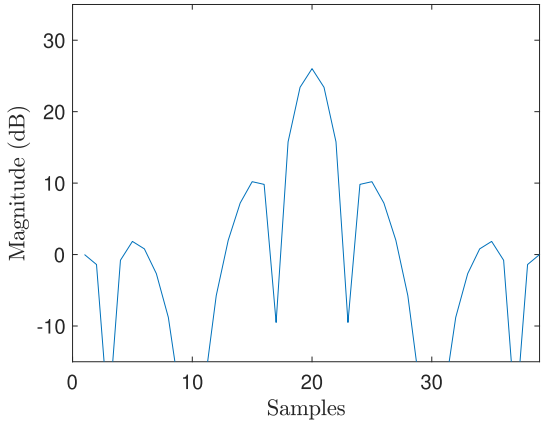


Figure 11.
Pulse autocorrelation.

RESULTS AND DISCUSSION

In this section, the simulation results of a complete radar are presented in a modular manner along with the analysis of each stage.

PULSE GENERATION

The process starts with the generation of an LFM pulse having real (I-part) and imaginary (Q-part) parts, as shown in Figure 10.

The *x*-axis in the figure displays the samples per pulse, while the *y*-axis displays the magnitude of the I and Q parts. It can be observed that the I-part and Q-part of the LFM pulse have a linearly increasing frequency and are 90°out of phase from each other. The autocorrelation of the pulse is shown in Figure 11. As there are 20 samples per pulse, the autocorrelation peaks at the 20th sample. Figure 12 displays the magnitude of each of the 1000 samples for a single pulse transmission and reception.

Since there are 1000 samples per PRI, the radar transmits for 20 samples and receives for 980 samples. The 1000 samples can be observed in Figure 12 that 20 samples are being transmitted, whereas the radar is in the receiving mode for the remaining 980 samples. In an actual radar, the cycle repeats after 1000 samples and continues for the operation of the radar. The 1000 samples corresponding to a single pulse form the fast-time axis, while the plotting of each subsequent pulse and its return (20 pulses in this case) forms the slow-time axis. Therefore, we have a fast-time axis representing the range sample axis consisting of 1000 samples per pulse and a

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

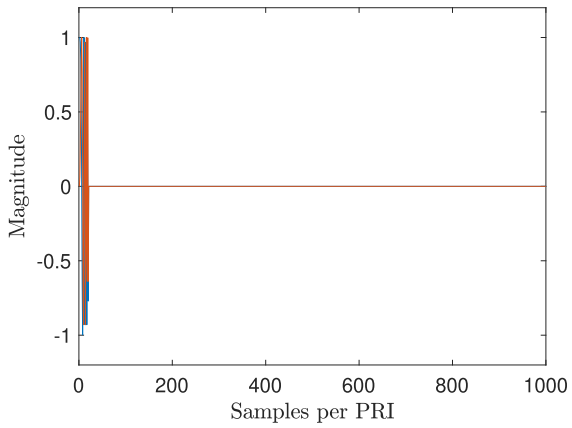


Figure 12.
Samples per PRI.

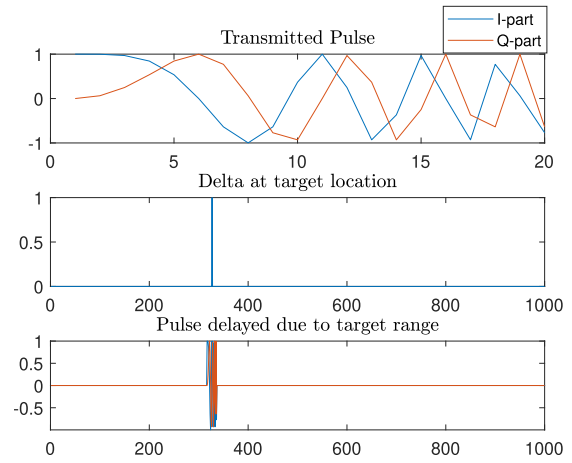


Figure 13.
Convolution of Dirac Delta function with a single target.

slow-time axis representing the pulse number axis. The fast-time axis is used to calculate the range of the target while the slow-time axis is utilized to compute the Doppler of the target.

TARGETS

The transmitted signal travels through the channel and strikes the targets, equivalent to convolution of the transmitted signal with a delayed impulse function as we have assumed the targets to be point scatterers for this simulation. The target simulation achieves the time delay by convolution of the transmitted pulse with the delayed delta function corresponding to each target. The resulting delayed pulse representing the target's range is shown in Figure 13 for the third target. We considered two types of losses that occur during the process; path loss caused by the wave travelling through free space and signal attenuation caused by RCS when the signal strikes the target. Additionally, an antenna gain has been modeled, which increases the strength of the signal intercepted by the radar.

For the five targets in Table 3, the convolution process with delayed Dirac delta function gives output in the form of a 2D matrix containing I and Q data for each pulse. Figure 14 shows the magnitude of the returns corresponding to the range samples (1000 in this case) for each pulse (20 in this case). The data contain echoes for 20 pulses in a sequence. The range sample axis is the fast-time axis, whereas the pulse number axis is the slow-time axis. The five targets, with the strength of signal dependent on the distance from the radar (due to path loss) and RCS of the target, can be seen clearly against noise. To visualize the data for a single pulse, we extract the information and display it in Figure 15.

Moreover, the range-sample axis is also converted to range axis, which shows that the radar's maximum range is 150 km, which is the same as theoretically computed. A part of these return signals reach the receiver on reflection. The receiver's goal is to detect the targets from the received signal and compute the target parameters. However, the received signal is weak since it has passed through a noisy channel and has been affected by path

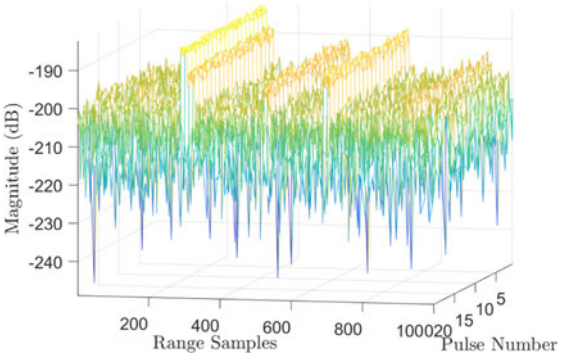


Figure 14.
Simulation of Targets IQ Data.

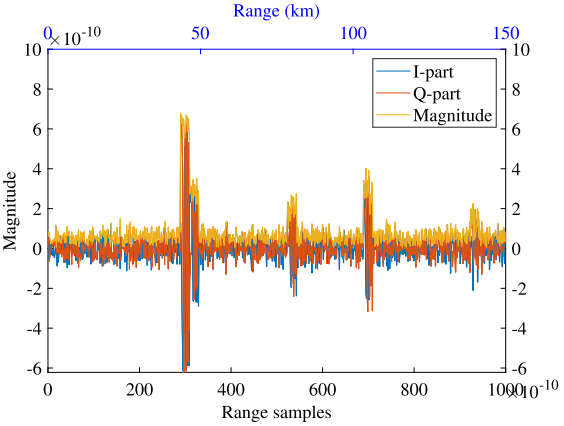


Figure 15.
Targets' IQ Data after convolution with generated pulse.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

loss. The attenuated received signal from a single pulse corrupted by noise is shown in Figure 16, where it is clear that the target returns from the corresponding ranges have been covered by noise. The receiver implements RSP algorithms to recover the target echoes from noise and extract targets' spikes from the background noise and clutter.

MATCHED FILTERING/PULSE COMPRESSION

The first process at the receiver front-end is matched filtering or pulse-compression that improves the range resolution and SNR. This step effectively finds the transmitted pulse in the received data. This process is achieved by convolution of received data with the time-reversed and complex conjugated transmitted pulse (to achieve correlation). The flipped conjugate transmitted pulse can be seen in Figure 17. Note that the flipped conjugate transmitted pulse is a linearly decreasing function of frequency, whereas the transmitted pulse is a linearly increasing function of frequency.

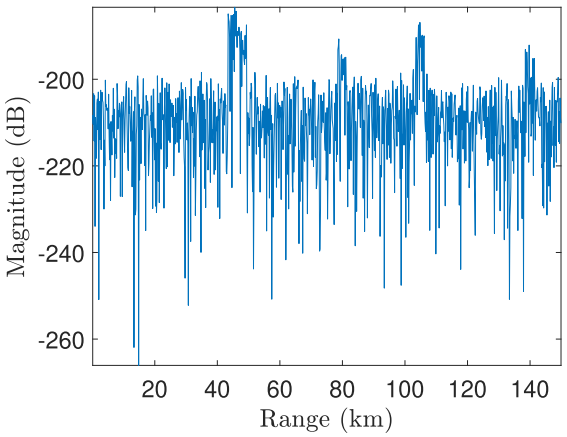


Figure 16.
Received signal before employing matched filtering.

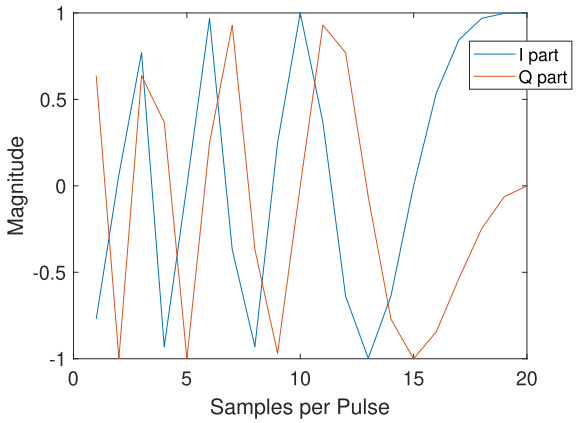


Figure 17.
I and Q parts of flipped conjugate transmitted pulse.

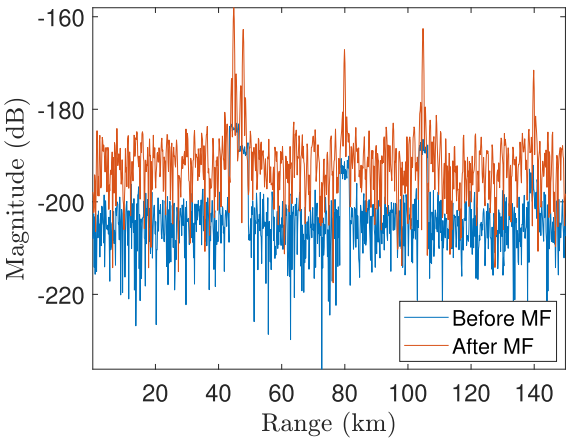


Figure 18.
Envelope of the signal before and after matched filtering.

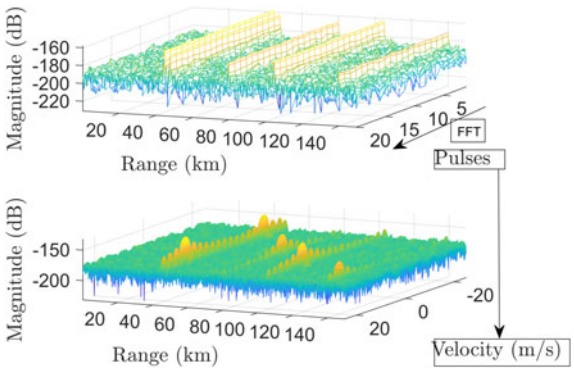


Figure 19.
Converting slow-time axis to velocity by taking FFT.

Figure 18 shows the envelope of the signals before and after matched filtering. Post matched filtering operation, the two overlapping targets, which were not resolved initially, are visible as two separate targets. Moreover, SNR improvement for the target peaks compared to the noise can also be seen after performing matched filtering. The next step is to apply Doppler Processing.

DOPPLER PROCESSING (FFT AND WINDOWING)

The simulated input and output of FFT-based Doppler processing block are shown in Figure 19.

Doppler processing has been implemented using the FFT algorithm to get the targets' velocity. The FFT is applied on the pulse number axis (the slow-time axis) as the phase difference between consecutive identical pulses can be used to detect the Doppler shift. Moving targets generate a time-varying spectral response in radars, which is captured by the FFT of the slow-time axis as the returns from identical sequential pulses are plotted on this axis. FFT of slow-time axis is used to convert time-domain samples to frequency-domain samples. FFT-based

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

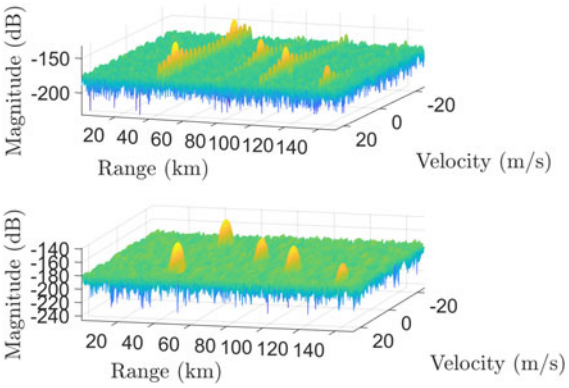


Figure 20.
Dolph–Chebyshev window implementation.

Doppler processing resolves targets according to their Doppler shift, independent of their range.

It can be seen that the slow-time axis is converted to the velocity axis with a range from -25 to 25 m/s (which is the maximum unambiguous Doppler velocity for the selected PRF), and all the five targets are visible at the correct simulated velocities. However, the peaks of the targets are surrounded by sidelobes. These sidelobes can be reduced by applying windowing operation. Therefore, using Dolph–Chebyshev windows on this signal (Figure 20) results in lower sidelobes but a broader main lobe. The widening of the main lobe with better suppression of sidelobes can be observed in Figure 20.

MTI FILTER

MTI is a high-pass filtering operation, and in this simulation, it is implemented using a three-pulse canceller. MTI filtering aims to remove all the clutter that lies at the low frequencies. However, the MTI filter also removes all the static targets, i.e., targets having zero velocity; therefore, this operation should suppress the stationary target simulated at a range of 80 km. Input and output of MTI Filter are shown in Figure 21(a) and (b), respectively. It is visible that MTI filter removes target at 80 km range with zero velocity along with the clutter.

Many modern radars have a moving target indication to separate moving targets from the clutter background since these radars must operate in environments with dense clutter. Doppler shift in frequency is utilized to distinguish moving targets in MTI radars, even when the echo signal from stationary targets is stronger than the target return. Fixed-target echoes or clutter are included in the same radar pulse as the target, but the fixed-target signals are not frequency shifted. Therefore, they are suppressed with the help of delay line cancellers, whereas moving targets are continued to be detected. Radars often incorporate both MTI filtering for clutter reduction and Doppler processing for Doppler spectrum analysis. Due to the linear nature of these processes, the sequence in which they are applied has no effect on the final Doppler spectrum. However, because clutter is the dominant component of the return signal, estimating Doppler prior to implementing MTI filtering may conceal possible targets traveling at near-zero velocities. In actual radars, the MTI process is thus implemented before the Doppler processing.

CONSTANT FALSE ALARM RATE

The CFAR's purpose is to establish the maximum power level at which any echo is likely to originate from a target. If the threshold is set too low, it will detect more echoes, increasing the risk of false alarms. When the threshold is set too high, few echoes are detected, and the number of missed detection increases. In most detectors, the threshold is frequently chosen to achieve a desired probability of false alarms. The CFAR calculates the detection

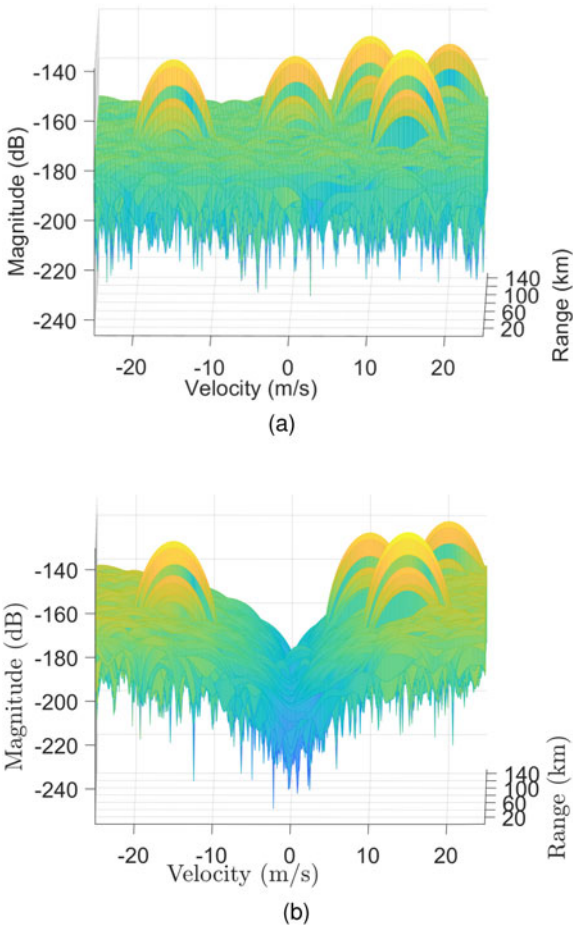


Figure 21.
MTI filter implementation. (a) Signal after FFT without MTI. (b) Signal after FFT with MTI.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

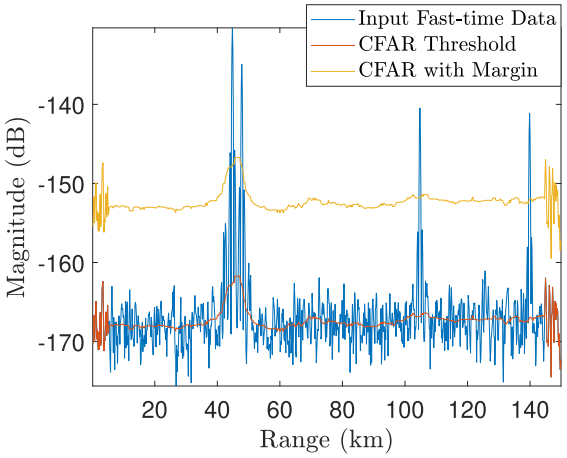


Figure 22.
Signal after OS CFAR.

threshold based on estimating the noise and interference levels present in the area, where target detection is done. The sample is called a cell in CFAR, where the cell under test (CUT) is often compared with other test cells. The noise power is estimated from neighboring cells and therefore, it is necessary to ignore adjacent cells which might contain a portion of the return signal. For this simulation, order statistics (OS CFAR) was used with 30 total cells and six guard cells on each side. The output of CFAR is shown in Figure 22 in which the fast-time data, CFAR threshold without and with margins can be observed.

DETECTED OUTPUT

Inputs crossing the CFAR threshold were considered as potential targets, and a peak-finding algorithm was used for targets detection. The output of the MATLAB console is shown in Figure 23.

The minor difference between the simulated and detected targets is the finite Range/Doppler resolution. The simulated and detected targets are compared in Table 4. This result shows that our receiver algorithm estimated the target parameters correctly.

COMPARISON OF RSP TECHNIQUES

The RSP techniques used in this tutorial are summarized in Table 5.

The effect of changing the pulse type (from LFM to rectangular), the number of FFT points (32 and 512 points), type of windows (Dolph–Chebyshev and Hanning window), MTI Filtering (two-pulse canceller and three-pulse canceller), and CFAR (CA CFAR and OS CFAR) on the radar signal are now presented. The effect of changing RCS on the signal return and Doppler ambiguities have also been visualized.

PULSE TYPE (LFM VERSUS RECTANGULAR SIGNAL)

If the pulse is changed from LFM waveform to a rectangular pulse, the radar cannot resolve close targets within the range resolution of the modeled radar (3 km). The simulation results shown in the Figure 24 demonstrate that while the LFM waveform was able to achieve resolution of the targets, the rectangular pulse is unable to achieve resolution of the targets.

This makes it clear that LFM waveforms have their advantage over rectangular waveforms when resolving closely spaced targets. Using LFM waveform allows equivalent sensitivity of a rectangular pulse, with a resolution improvement proportional to the bandwidth of the LFM waveform.

CFAR COMPARISON

The output of OS CFAR is shown in Figure 22. CA CFAR has also been implemented (Figure 25), which is based on a moving average. The CA CFAR implemented in this tutorial is based on the assumption of uniform clutter in the reference window. When the returns from a strong target are close to the returns from a weaker target, CA CFAR performs poorly because it masks the targets. On the other hand, because OS CFAR does not conceal a weak target near a strong target return, its performance is superior in this scenario. As a result, the type of CFAR implemented is dependent on the target type for which the radar is built.

UNDERSTANDING DOPPLER AMBIGUITIES

To understand Doppler ambiguities, the target at 105 km range is simulated at a velocity of 40 m/s, and it wraps to −10 m/s since the unambiguous Doppler range is [−25 25] m/s for the selected PRF, which is exactly what can be seen in Figure 26. As a result, it is vital to tailor the radar characteristics to the application

```
Target#1                Range=44.8 Km              Velocity=15.03 m/s
Target#2                Range=47.8 Km              Velocity=-15.03 m/s
Target#3                Range=104.8 Km              Velocity=10.05 m/s
Target#4                Range=139.9 Km              Velocity=19.91 m/s
```

Figure 23.
MATLAB console output.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

Table 4.

Detected Versus Simulated Target Parameters			
	Simulated Range/ Velocity	Detected Range/ Velocity	Detection
Target-1	45 / 15	44.8 / 15.03	Yes
Target-2	47.9 / −15	47.8 / −15.03	Yes
Target-3	80 / 0	- / -	No since it was a static target
Target-4	105 / 10	104.8 / 10.05	Yes
Target-5	140 / 20	139.9 / 20.01	Yes

Table 5.

Summary of RSP Techniques for RSP Operations		
S No	Radar Operation	RSP Technique
1	Target Range Simulation	Convolution
2	Matched Filtering	Convolution
3	Pulse Compression	Autocorrelation
4	Doppler Processing	FFT
5	Doppler side lobe suppression	Windowing
6	MTI	High pass filtering
7	CFAR	Modified Moving Average / Median Filtering

for which they are being designed. It should be capable of determining the maximum velocity required for the application. Additionally, radars with staggered PRFs are already in use, allowing for simultaneous enhancement of the radar's maximum unambiguous range and velocity.

WINDOWS COMPARISON

Windowing operation is used to suppress sidelobes produced by FFT operation along the Doppler axis. To illustrate the effects of different windows on the targets, we have used two different windows, i.e., Dolph–Chebyshev

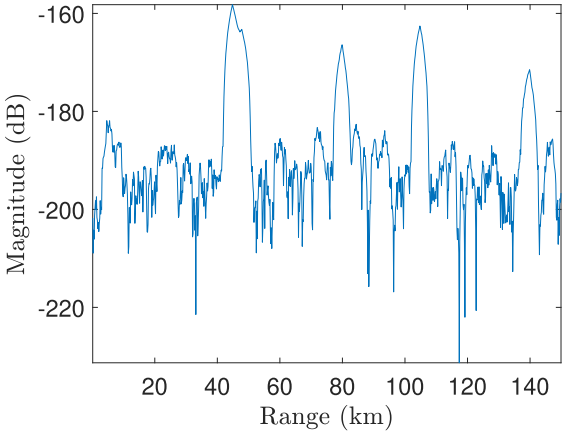


Figure 24.
Matched filter output for a rectangular pulse.

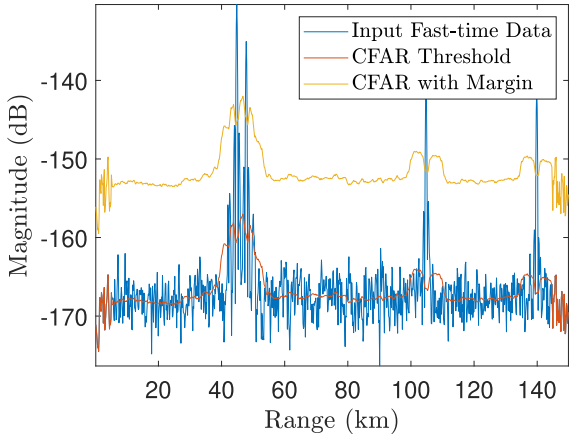


Figure 25.
CA CFAR implementation.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

and Hanning window. The impact of using Dolph–Chebyshev windows and Hanning windows on the radar Doppler axis (the slow-time axis) are shown in Figure 27(a) and (b), respectively. It can be seen that Dolph–Chebyshev windows suppress the sidelobes better but have a broader main lobe compared to the Hanning window, whereas for the Hanning window, the main lobe is narrower than the Dolph–Chebyshev window. However, the sidelobes' suppression is not as good as the Dolph–Chebyshev window in this case.

There is no universal method for determining the window function to use. The use of the Windowing operation depends on the type of application the radar is being used for. In case of an application where there are strong interfering signals near the signal of interest, choose a window function with a low maximum sidelobe level (SLL). For example, detecting a target in the presence of jammer where sidelobe suppression is important, the Dolph–Chebyshev will be the preferred window as it has a lower noise threshold than the other windows. The main lobe will be broader in this case. In applications where a narrower main lobe is required, like detecting a small car next to a large truck, Hanning Window should be implemented as it has a narrower main lobe. Hence, selection of Window is a design tradeoff, where lower sidelobes can be traded for a wider main lobe.

MTI FILTER

MTI filter suppresses returns at zero frequency since the static Doppler contains all the clutter. For the same scenario, performance of two MTI filters, the two-pulse canceller and the three-pulse canceller was evaluated. Figure 28(a) and (b) illustrates how a two-pulse canceller and a three-pulse canceller suppress a static target (zero velocity), respectively. It can be observed that a three-pulse canceller suppresses the stationary target more effectively than a two-pulse canceller by suppressing the signals at near-zero Doppler frequencies.

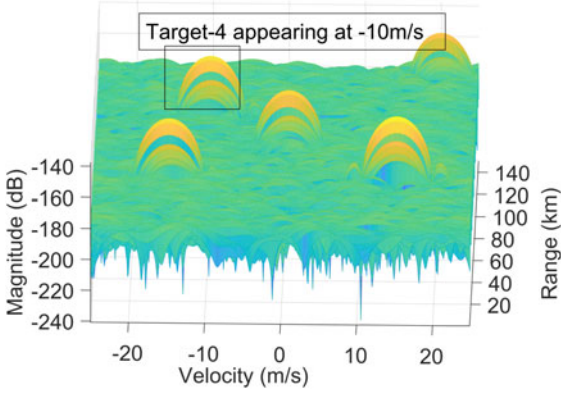


Figure 26.
Doppler ambiguity for a particular PRF.

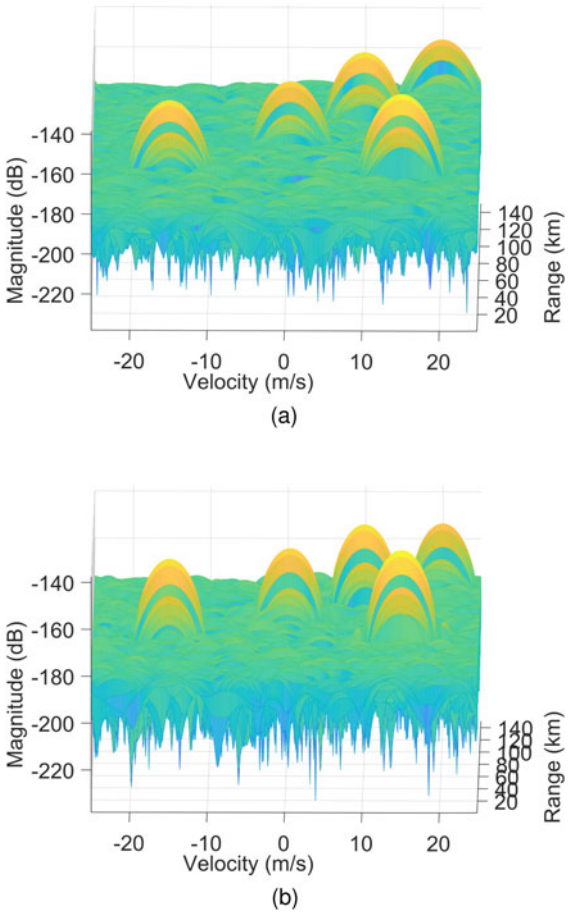


Figure 27.
Effect of Windowing operation on slow-time axis. Dolph–Chebyshev Window. (b) Hanning window.

NUMBER OF FFT POINTS

The effect of using different number of FFT points on the slow-time axis to get the desired velocity of the target will be in terms of interpolation of the velocity axis. A higher number of FFT point would allow for better velocity resolution in comparison with low number of FFT points. To demonstrate this effect, 32 and 512 point FFT of the slow-time axis is carried out. Figure 29(a) and (b) shows the Doppler domain plot for an FFT size equal to 32 and 512, respectively. It can be observed that the velocity resolution for 512 point FFT is better than 32 point FFT. However, this comes at the cost of utilizing more hardware resources.

EFFECT OF RCS

For targets with greater RCS, the strength of targets' return would be greater than targets with lower RCS. To

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

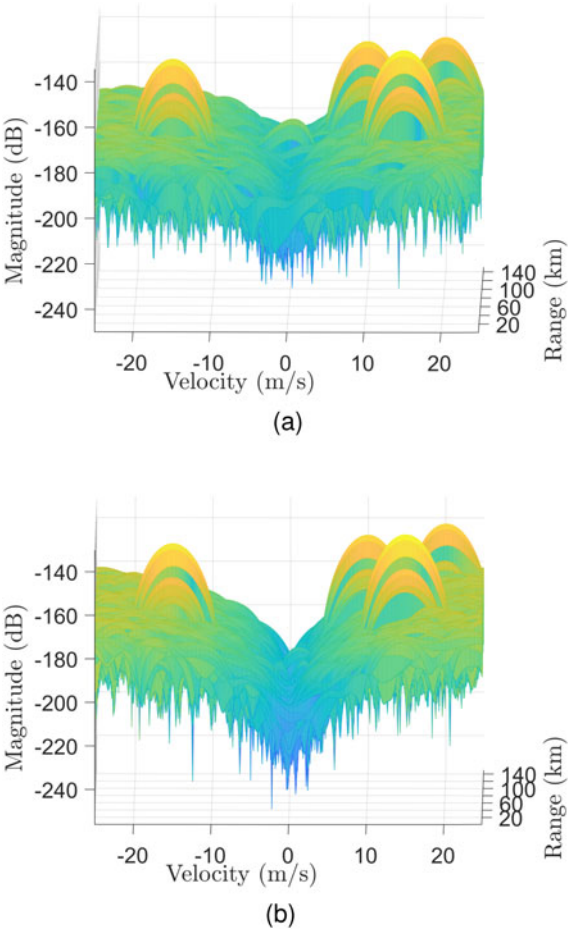


Figure 28.
Effect of MTI Filter on clutter suppression. (a) Two-pulse canceller. (b) Three-pulse canceller.

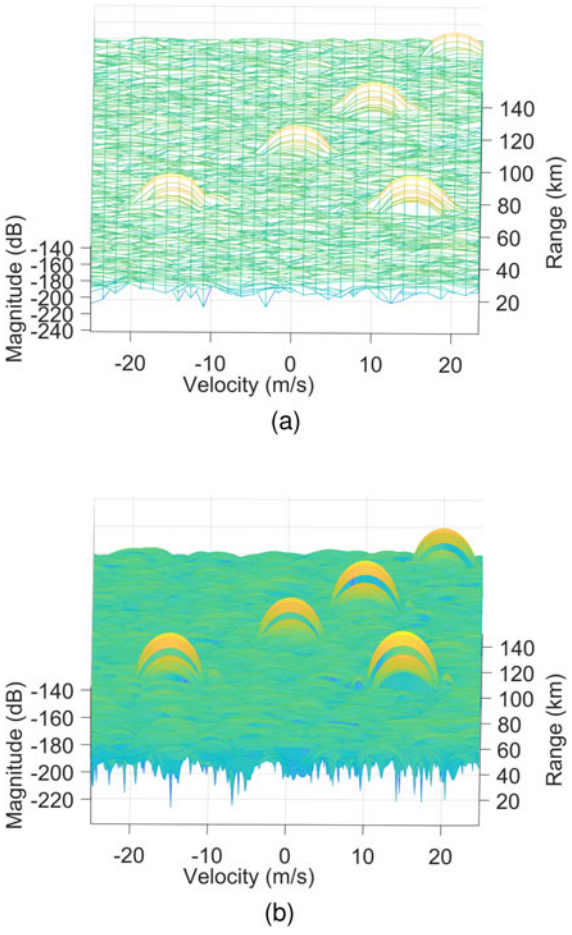


Figure 29.
Effect of FFT points on resolution. (a) 40 point FFT. (b) 512 point FFT.

demonstrate the effect of different RCS on the radar performance, the RCS for the last two targets was increased to 50 m^2 and 100 m^2, respectively [see Figure 30(a)], from 5 m^2 and 8 m^2 [see Figure 30(b)]. The static target is suppressed for any value of RCS as it is stationary, i.e., it is present at zero Doppler. Targets having a greater RCS are detected more accurately than targets with a smaller RCS. As the RCS decreases, targets with returns near the noise floor may be overlooked entirely. This is the underlying premise of stealth aircraft design, where their lower RCS allows them to fly undetected by a conventional radar. They may, however, be noticed when they get closer to the radar.

Considering the complexity of RCS phenomenology owing to various characteristics such as geometry, size, or orientation of the objects, and frequency and polarization of the radar antenna, simplifications are employed to estimate the RCS of targets. For example, the target has been deemed to have a single scattering point. In addition, the return is considered to fit within a single range bin, thereby eliminating the possibility of numerous scattering locations or some backscatter spread. The target has also been assumed to be in the far field to consider the plane wave case for simplifications.

CONCLUSION AND FUTURE WORK

Radar system designers face many conflicting design issues while developing a radar system. The PRF, the pulsewidth, the operating frequency, the antenna characteristics, and the application for which the radar will be employed all play a significant impact in determining the parameters of a target such as range, velocity, and bearing that can be detected by a radar. It is vital to understand the processes involved in developing any radar system, whether the intended system is being developed to detect 2D/3D parameters of a target [16] or conduct reconfigurable intelligent surface (RIS)-assisted radar sensing [63]. Once the angle is determined or the effects of RIS on the transmission and reception paths are taken into account, the following

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

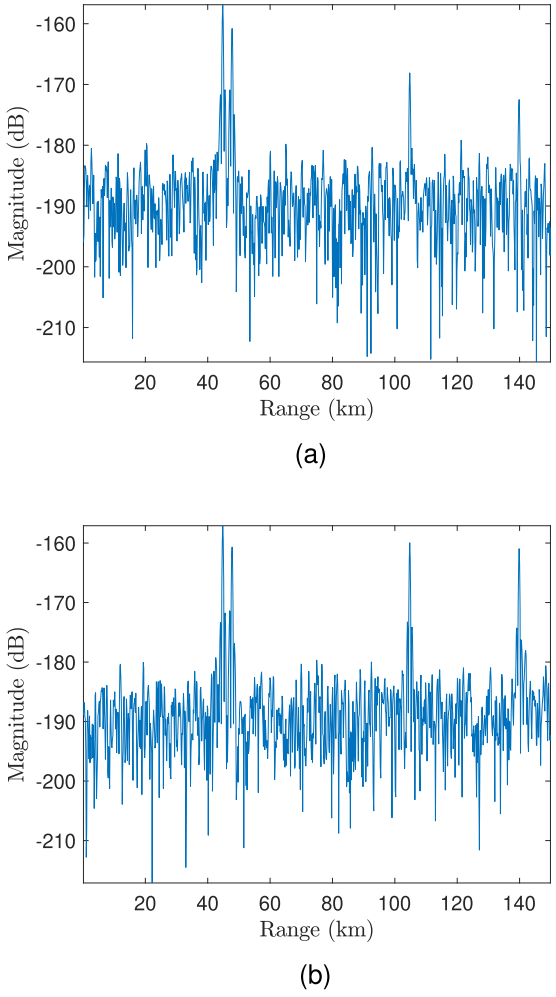


Figure 30.
Effect of RCS on return signal. (a) Decreased RCS for the last two targets. (b) Increased RCS for the last two targets.

steps are invariably applying range-Doppler processing, clutter suppression procedures, and CFAR techniques [64]. This tutorial outlines a modular framework for RSP by breaking the radar system into its component subsystems and then sharing the output of each subsystem to provide a holistic view of its operation along with the involved RSP techniques. The understanding of RSP techniques in radar system operations allows the designer to visualize the radar signals as they travel through the various subsystems of the radar. The selection of the pulse type, window type, number of FFT points, CFAR approach, and peak-detecting algorithm all affect the radar system's performance. For this tutorial, the RSP techniques used in radar operations have been demonstrated using a 2D S-band radar system, with the signal displayed before and after each operation phase. Additionally, we analyzed the effect of each presented RSP technique used in radar operations on target detection. The tutorial includes the code to aid the reader in learning how each radar subsystem operates. We have shown the pre-and postprocessed signals at each stage to help understand the process's basic concepts and math. Further research to identify how a radar system operates in the presence of RIS in the sensing architecture to detect target parameters can be conducted. This scenario finds utility in a wide variety of radar applications. The operation of an architecture consisting of a 3D radar system and RIS can be explained in detail, as such an exposition would benefit the entire radar community.

ACKNOWLEDGMENTS

The authors would like to thank Ministry of Higher Education Malaysia for Fundamental Research Grant Scheme with Project Code FRGS/1/2022/TK07/USM/02/14 for permitting them to carry out this research.

REFERENCES

[1] M. Skolnik, *Introduction to Radar Systems*. New York, NY, USA: McGraw-Hill, 2001.

[2] S. Soclof et al., *The Electronics Handbook*, 2nd ed. New York, NY, USA: Taylor and Francis, 2005.

[3] S. M. Patole, M. Torlak, D. Wang, and M. Ali, "Automotive radars: A review of signal processing techniques," *IEEE Signal Process. Mag.*, vol. 34, no. 2, pp. 22–35, Mar. 2017.

[4] P. S. Diao, T. Alves, B. Poussot, and S. Azarian, "A review of radar detection fundamentals," *IEEE Aerosp. Electron. Syst. Mag.*, 2022, doi: 10.1109/MAES.2022.3177395.

[5] M. A. Richards, *Fundamentals of Radar Signal Processing*. New York, NY, USA: McGraw-Hill Professional, 2005.

[6] B. Cantrell et al., "Development of a digital array radar (DAR)," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 17, no. 3, pp. 22–27, Mar. 2002.

[7] C. Fulton, M. Yeary, D. Thompson, J. Lake, and A. Mitchell, "Digital phased arrays: Challenges and opportunities," *Proc. IEEE*, vol. 104, no. 3, pp. 487–503, Mar. 2016.

[8] J. S. Herd and M. D. Conway, "The evolution to modern phased array architectures," *Proc. IEEE*, vol. 104, no. 3, pp. 519–529, Mar. 2016.

[9] W. Melvin, "A STAP overview," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 19, no. 1, pp. 19–35, Jan. 2004.

[10] A. Barros Cardoso da Silva, S. K. Joshi, S. V. Baumgartner, F. Q. de Almeida, and G. Krieger, "Phase correction for accurate DOA angle and position estimation of ground-moving targets using multi-channel airborne radar," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 4021605.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

[11] S. Gogineni and A. Nehorai, "Monopulse MIMO radar for target tracking," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 47, no. 1, pp. 755–768, Jan. 2011.

[12] A. M. Ahmed, U. S. K. P. M. Thanthrige, A. E. Gamal, and A. Sezgin, "Deep learning for DOA estimation in MIMO radar systems via emulation of large antenna arrays," *IEEE Commun. Lett.*, vol. 25, no. 5, pp. 1559–1563, May 2021.

[13] M. Younis, F. Q. de Almeida, M. Villano, S. Huber, G. Krieger, and A. Moreira, "Digital beamforming for space-borne reflector-based synthetic aperture radar, part 1: Basic imaging modes," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 3, pp. 8–25, 2021.

[14] J. Sowers, N. Byer, B. Edward, D. McPherson, S. Weinreb, and F. Rucky, "Electronically steered, receive monopulse, active phased array at 94 ghz," in *Proc. IEEE MTT-S Int. Microw. Symp. Dig.*, vol. 3, 1996, pp. 1581–1584.

[15] L. Di Palma, A. Clemente, L. Dussopt, R. Sauleau, P. Potier, and P. Pouliguen, "Radiation pattern synthesis for monopulse radar applications with a reconfigurable transmit array antenna," *IEEE Trans. Antennas Propag.*, vol. 64, no. 9, pp. 4148–4154, Sep. 2016.

[16] G. Babur, G. O. Manokhin, A. A. Geltser, and A. A. Shibelgut, "Low-cost digital beamforming on receive in phased array radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 3, pp. 1355–1364, Jun. 2017.

[17] R. Takahashi, K. Hirata, T. Hara, and A. Okamura, "Derivation of monopulse angle accuracy for phased array radar to achieve Cramer-Rao lower bound," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2012, pp. 2569–2572.

[18] T. Tirer and O. Bialer, "Effective approximate maximum likelihood estimation of angles of arrival for non-coherent sub-arrays," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, 2020, pp. 4557–4561.

[19] U. Nickel, "Overview of generalized monopulse estimation," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 21, no. 6, pp. 27–56, Jun. 2006.

[20] R. M. Davis and R. L. Fante, "A maximum-likelihood beamspace processor for improved search and track," *IEEE Trans. Antennas Propag.*, vol. 49, no. 7, pp. 1043–1053, Jul. 2001.

[21] D. Efstathiou, L. Fridman, and Z. Zvonar, "Recent developments in enabling technologies for software defined radio," *IEEE Commun. Mag.*, vol. 37, no. 8, pp. 112–117, Aug. 1999.

[22] M. Mody, H. Hariyani, A. Balagopalakrishnan, J. Jones, A. Jayaraj, and Y. Prithvishankar, "GPU assist using DSP pre-processor," in *Proc. IEEE Int. Conf. Electron. Comput. Commun. Technol.*, 2020, pp. 1–4.

[23] C. Cheung, R. Shah, and M. Parker, "Time delay digital beamforming for wideband pulsed radar implementation," in *Proc. IEEE Int. Symp. Phased Array Syst. Technol.*, 2013, pp. 448–455.

[24] X. Xin, Y. Qinghong, S. Zhaolin, L. Qingjiang, W. Yinan, and L. Wenli, "The realization of digital beamforming based on FPGA and dsp," in *Proc. Int. Conf. Intell. System Des. Eng. Appl.*, vol. 2, 2010, pp. 713–716.

[25] S. H. Talisa, K. W. O'Haver, T. M. Comberiate, M. D. Sharp, and O. F. Somerlock, "Benefits of digital phased array radars," *Proc. IEEE*, vol. 104, no. 3, pp. 530–543, Mar. 2016.

[26] W. Hao and Z. Hong-zhong, "Modeling and simulation of a full coherent LFM pulse radar system based on simulink," in *Proc. 2nd Int. Conf. Meas. Inf. Control*, vol. 1, 2013, pp. 495–499.

[27] D. K. Barton, "Simple procedures for radar detection calculations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES- 5, no. 5, pp. 837–846, Sep. 1969.

[28] D. K. Barton, *Radar Equations for Modern Radar*. Norwood, MA, USA: Artech House, 2013.

[29] C. A. Balanis, *Antenna Theory: Analysis and Design*. Hoboken, NJ, USA: Wiley, 2015.

[30] P. Swerling, "Probability of detection for fluctuating targets," *IRE Trans. Inf. Theory*, vol. 6, no. 2, pp. 269–308, Apr. 1960.

[31] P. Swerling, "Radar probability of detection for some additional fluctuating target cases," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 33, no. 2, pp. 698–709, Apr. 1997.

[32] J. R. Klauder, A. Price, S. Darlington, and W. J. Albersheim, "The theory and design of chirp radars," *Bell Syst. Tech. J.*, vol. 39, no. 4, pp. 745–808, 1960.

[33] V. Jain and W. D. Blair, "Filter design for steady-state tracking of maneuvering targets with LFM waveforms," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 45, no. 2, pp. 765–773, Apr. 2009.

[34] B. R. Mahafza and A. Z. Elsherbeni, *Simulations for Radar Systems Design*. London, U.K.: Chapman&Hall/CRC, 2004, pp. 6–12.

[35] M. Xia, W. Su, and H. Gu, "The research on Doppler ambiguity for dual pulse repetition frequencies radar," in *Proc. CIE Int. Conf. Radar*, 2016, pp. 1–4.

[36] O. Çulha and Y. Tanik, "The detection of fluctuating targets in case of range migration," in *Proc. 25th Signal Process. Commun. Appl. Conf.*, 2017, pp. 1–4.

[37] S. M. Kay and S. L. Marple, "Spectrum analysis—A modern perspective," *Proc. IEEE*, vol. 69, no. 11, pp. 1380–1419, Nov. 1981.

[38] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall Press, 2009.

[39] H. Rakshit and M. A. Ullah, "A comparative study on window functions for designing efficient FIR filter," in *Proc. 9th Int. Forum Strategic Technol.*, 2014, pp. 91–96.

[40] A. Zverev, "Digital MTI radar filters," *IEEE Trans. Audio Electroacoust.*, vol. 16, no. 3, pp. 422–432, Sep. 1968.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.

[41] R. S. Raghavan, "CFAR detection in clutter with a Kronecker covariance structure," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 2, pp. 619–629, Apr. 2017.

[42] G. M. Dillard, "Mean-level detection of nonfluctuating signals," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-10, no. 6, pp. 795–799, Nov. 1974.

[43] M. Weiss, "Analysis of some modified cell-averaging CFAR processors in multiple-target situations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-18, no. 1, pp. 102–114, Jan. 1982.

[44] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-19, no. 4, pp. 608–621, Jul. 1983.

[45] M. Shor and N. Levanon, "Performances of order statistics CFAR," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 27, no. 2, pp. 214–224, Mar. 1991.

[46] L. Wang, J. Tang, and Q. Liao, "A study on radar target detection based on deep neural networks," *IEEE Sens. Lett.*, vol. 3, no. 3, Mar. 2019, Art no. 7000504.

[47] Y. Liu, S. Zhang, J. Suo, J. Zhang, and T. Yao, "Research on a new comprehensive CFAR (comp-CFAR) processing method," *IEEE Access*, vol. 7, pp. 19401–19413, 2019.

[48] C. Wang, J. Wang, and X. Zhang, "Automatic radar waveform recognition based on time-frequency analysis and convolutional neural network," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2017, pp. 2437–2441.

[49] S. Blake, "OS-CFAR theory for multiple targets and nonuniform clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 24, no. 6, pp. 785–790, Nov. 1988.

[50] R. G. Zefreh, M. R. Taban, M. M. Naghsh, and S. Gazor, "Robust CFAR detector based on censored harmonic averaging in heterogeneous clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 3, pp. 1956–1963, Jun. 2021.

[51] Y. Wang, W. Xia, and Z. He, "CFAR knowledge-aided radar detection with heterogeneous samples," *IEEE Signal Process. Lett.*, vol. 24, no. 5, pp. 693–697, May 2017.

[52] S. Wunsch, J. Fink, and F. K. Jondral, "Improved detection by peak shape recognition using artificial neural networks," in *Proc. IEEE 82nd Veh. Technol. Conf.*, 2015, pp. 1–5.

[53] R. Wang, J. Li, Y. Duan, H. Cao, and Y. Zhao, "Study on the combined application of CFAR and deep learning in ship detection," *J. Indian Soc. Remote Sens.*, vol. 46, no. 9, pp. 1413–1421, 2018.

[54] M. Kang, X. Leng, Z. Lin, and K. Ji, "A modified faster R-CNN based on CFAR algorithm for SAR ship detection," in *Proc. IEEE Int. Workshop Remote Sens. Intell. Process.*, 2017, pp. 1–4.

[55] A. Lulu and B. G. Mobasseri, "Phase matching of coincident pulses for range-Doppler estimation of multiple targets," *IEEE Signal Process. Lett.*, vol. 26, no. 1, pp. 199–203, Jan. 2019.

[56] S. M. D. Rizvi, S. Ahmad, K. Khan, A. Hasan, and A. Masood, "Deep learning approach for fixed and rotary-wing target detection and classification in radars," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 37, no. 3, pp. 32–42, Mar. 2022.

[57] Y.-P. Cheng and J.-Y. Nieh, "Precise range and doppler estimation of multi-nonstationary targets by LFM pulse-doppler radars," in *Proc. Int. Conf. Radar, Antenna, Microw., Electron., Telecommun.*, 2018, pp. 21–26.

[58] I. Intyas, R. Hasanah, M. R. Hidayat, B. Hasanah, A. B. Suskmono, and A. Munir, "Improvement of radar performance using LFM pulse compression technique," in *Proc. Int. Conf. Elect. Eng. Inform.*, 2015, pp. 302–307.

[59] Z. Li, N. Tai, C. Wang, and N. Yuan, "A study on blanket noise jamming to LFM pulse compression radar," in *Proc. IEEE Int. Conf. Signal Process. Commun. Comput.*, 2017, pp. 1–5.

[60] B. Sundaram and P. N. Shastry, "A novel electronically tunable active duplexer for wireless transceiver applications," *IEEE Trans. Microw. Theory Technol.*, vol. 54, no. 6, pp. 2584–2592, Jun. 2006.

[61] W. L. Stutzman, "Estimating directivity and gain of antennas," *IEEE Antennas Propag. Mag.*, vol. 40, no. 4, pp. 7–11, Aug. 1998.

[62] C.-T. Tai and C. Pereira, "An approximate formula for calculating the directivity of an antenna," *IEEE Trans. Antennas Propag.*, vol. 24, no. 2, pp. 235–236, Mar. 1976.

[63] S. Buzzi, E. Grossi, M. Lops, and L. Venturino, "Radar target detection aided by reconfigurable intelligent surfaces," *IEEE Signal Process. Lett.*, vol. 28, pp. 1315–1319, 2021.

[64] A. Aubry, A. De Maio, and M. Rosamilia, "RIS-aided radar sensing in N-LOS environment," in *Proc. IEEE 8th Int. Workshop Metrol. AeroSp.*, 2021, pp. 277–282.



Salman Liaquat (Graduate Student Member, IEEE) received the B.E. degree in avionics engineering from National University of Sciences and Technology, Risalpur, Pakistan, in 2010, and the M.S. degree in avionics engineering from Air University, Islamabad, Pakistan, in 2019. He is currently working toward the Ph.D. degree in electrical and electronic engineering with Universiti Sains Malaysia, Penang, Malaysia. His research interests include signal processing for communications, mainly focusing in the area of radar signal processing. He is also interested in reconfigurable intelligent surfaces and their applications. He is a member of Royal Aeronautical Society, U.K.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.



Nor Muzlifah Mahyuddin (Member, IEEE) received the B.Eng. degree from Universiti Teknologi Malaysia, Malaysia, in 2005, the M.Sc. degree from Universiti Sains Malaysia, Malaysia, in 2006, and the Ph.D. degree from Newcastle University, Newcastle upon Tyne, U.K., in 2011. She is currently an Associate Professor with the School of Electrical and Electronic Engineering, Universiti Sains Malaysia, Penang, Malaysia. Her research interests are in the field of art of miniaturization and low power in RF and microwave engineering, and digital integrated circuit.



Ijaz Haider Naqvi (Senior Member, IEEE) received the B.Sc. degree in electrical engineering from the University of Engineering and Technology Lahore, Pakistan, in 2003, the master's degree in radio communications from SUPELEC Paris, France, in 2006, and the Ph.D. degree in electronics and telecommunications from IETR-INSA Rennes, France, in 2009. He is currently an Associate Professor with the School of Science and Engineering, Lahore University of Management Sciences, Lahore, Pakistan. He has several years of research experience in the wireless communications and wireless sensor networks. His current research interests include 5G networks, millimeter wave wireless systems for 5G and 6G telecommunication and multiantenna radar systems for civilian applications.

Authorized licensed use limited to: ULAKBIM UASL ISTANBUL TEKNIK UNIV. Downloaded on July 21,2026 at 05:30:45 UTC from IEEE Xplore. Restrictions apply.